

Homework Submission

Student Name
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Problem Statement

Evaluate the indefinite integral

$$\int x^2 dx.$$

Key Principles and Formulas

The power rule for integration states that for any real number $n \neq -1$,

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C,$$

where C is the constant of integration. The rule is derived by reversing the power rule for differentiation, $\frac{d}{dx}x^k = kx^{k-1}$.

Detailed Step-by-Step Solution

First, write down the general power rule for integration.

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C \quad (n \neq -1)$$

Now apply the rule with the specific exponent $n = 2$.

$$\begin{aligned} \int x^2 dx &= \frac{x^{2+1}}{2+1} + C \\ &= \frac{x^3}{3} + C. \end{aligned}$$

Final Answer

$$\boxed{\frac{x^3}{3} + C}$$

Study Notes and Key Takeaways

- Remember to increase the exponent by one **before** dividing by the new exponent.
- The constant of integration C is essential for indefinite integrals; omitting it leads to an incomplete antiderivative.
- The power rule does **not** apply when the exponent is -1 ; in that case the integral is $\int x^{-1}dx = \ln|x| + C$.